Record Consultation – System Operation Contracts

Contract CO8 – Search Arrived Appointments

Field	Answer
Operation:	searchArrivedAppointments(searchCriteria)
Use Case:	Record Consultation
Preconditions:	Healthcare Professional is authenticated and authorized to search appointments. searchCriteria contains at least one of: insurance number, date of birth, phone number, or partial name
Success Postconditions:	System returns the appointments of the current day, assigned to the Healthcare Professional, that match the search criteria and whose arrival_status is Arrived. No domain objects are modified
Failure Postconditions:	No matching appointments are returned. No domain objects are modified

Contract CO9 – Start Consultation

Field	Answer
Operation:	startConsultation(appointmentId)
Use Case:	Record Consultation
Preconditions:	Healthcare Professional is authenticated and authorized to conduct consultations. A validly formatted appointment identifier has been provided
Success Postconditions:	The Appointment is confirmed as assigned to the Healthcare Professional, scheduled for the current day, marked as Arrived, and without an existing Consultation A new Consultation instance is created (instance creation) The Consultation is associated with the Appointment and the Healthcare Professional (association formed)
Failure Postconditions:	No Consultation instance is created. No domain objects are modified. The system returns a failure indication

Contract CO10 – Record Clinical Information

Field	Answer
Operation:	recordClinicalInformation(consultationId, clinicalInfo)
Use Case:	Record Consultation
Preconditions:	Healthcare Professional is authenticated and authorized to record clinical information. A started Consultation exists for the provided identifier. clinicalInfo contains at least one of: observations, diagnoses, treatment recommendations, follow-up requirements

Success Postconditions:	A new ClinicalRecord instance is created (instance creation) The ClinicalRecord's observations, diagnoses, treatment_recommendations and followup_requirements are set from clinicalInfo (attribute modification) The ClinicalRecord is associated with the Consultation (association formed) A new ActivityRecord instance is created to document the access to and recording of clinical information, and associated with the responsible user (instance creation, association formed)
Failure Postconditions:	No ClinicalRecord instance is created. No ActivityRecord is created. The system returns a failure indication

Contract CO11 – Issue Prescription

Field	Answer
Operation:	issuePrescription(consultationId, medications)
Use Case:	Record Consultation
Preconditions:	Healthcare Professional is authenticated and authorized to issue prescriptions. A started Consultation exists for the provided identifier. medications contains at least one valid Medication
Success Postconditions:	A new Prescription instance is created (instance creation) The Prescription is associated with the issuing Healthcare Professional and the Consultation (association formed) The Prescription is associated with each prescribed Medication (association formed) A new ActivityRecord instance is created to document the prescription (instance creation, association formed)
Failure Postconditions:	No Prescription instance is created. No associations are formed. No ActivityRecord is created. The system returns a failure indication

Contract CO12 – End Consultation

Field	Answer
Operation:	endConsultation(consultationId)
Use Case:	Record Consultation
Preconditions:	Healthcare Professional is authenticated and authorized to conduct consultations. A started Consultation exists for the provided identifier
Success Postconditions:	The Consultation is marked as Completed (attribute modification) The Appointment's status is changed to Completed (attribute modification) A new ActivityRecord instance is created to document the consultation completion (instance creation, association formed)
Failure Postconditions:	The Consultation and Appointment statuses remain unchanged. No ActivityRecord is created. The system returns a failure indication